All Descendants (Query 01)

Graph	Query	Parameters	Apache AGE					Kuzu					Neo4j				
			B	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P
F40	Q_{desc}	Root	9.6	1.4	1.4	6.7x	6.9x	3.0	1.9	1.9	1.6x	1.6x	2.0	2.0	2.3	0.99x	0.88x
F40	Q_{desc}	Leaf parent H	8.8	1.4	1.4	6.3x	6.4x	2.7	1.4	1.3	1.9x	2.1x	1.3	1.6	1.5	0.86x	0.87x
F40	Q_{desc}	Leaf parent	8.9	1.4	1.4	6.4x	6.4x	2.7	1.9	1.7	1.4x	1.6x	1.3	1.5	1.7	0.86x	0.78x
F1K	Q_{desc}	Root	440	2.3	2.3	190x	190x	6.3	3.1	2.5	2.1x	2.5x	30	28	41	1.0x	0.71x
F1K	Q_{desc}	Leaf parent H	13	2.3	2.3	5.9x	5.9x	4.2	1.9	2.2	2.2x	1.9x	1.9	2.8	2.3	0.68x	0.85x
F1K	Q_{desc}	Leaf parent	13	1.9	1.8	6.7x	7.1x	3.6	2.0	2.0	1.7x	1.8x	1.8	1.7	1.9	1.0x	0.95x
NT10	Q_{desc}	Root	9.3	1.3	1.3	7.0x	6.9x	2.8	2.0	1.7	1.4x	1.6x	2.2	2.1	2.3	1.0x	0.96x
NT10	Q_{desc}	Leaf parent H/L	9.3	1.3	1.3	7.0x	6.9x	2.6	1.7	1.5	1.5x	1.7x	1.4	1.7	1.6	0.83x	0.88x
NT100	Q_{desc}	Root	20	1.6	1.5	13x	13x	3.9	1.7	1.8	2.2x	2.2x	11	13	14	0.86x	0.84x
NT100	Q_{desc}	Leaf parent H	9.9	1.5	1.4	6.8x	7.0x	3.1	1.8	1.6	1.7x	1.9x	2.2	2.4	2.3	0.91x	0.94x
NT100	Q_{desc}	Leaf parent	8.8	1.4	1.4	6.1x	6.2x	2.9	1.5	1.7	1.9x	1.8x	1.5	1.6	1.8	0.95x	0.83x
NT1K	Q_{desc}	Root	1.1K	2.8	2.8	400x	390x	10	2.0	2.0	5.2x	4.9x	58	46	48	1.3x	1.2x
NT1K	Q_{desc}	Leaf parent H	16	2.3	2.3	6.7x	6.8x	2.9	1.5	1.7	1.9x	1.7x	1.3	1.4	1.5	0.93x	0.90x
NT1K	Q_{desc}	Leaf parent	12	2.3	2.3	5.4x	5.5x	3.8	1.7	1.8	2.2x	2.1x	1.2	1.2	1.3	0.99x	0.90x
NT10K	Q_{desc}	Root	110K	16	16	6.9Kx	6.9Kx	100	11	9.2	9.5x	11x	420	490	530	0.85x	0.78x
NT10K	Q_{desc}	Leaf parent H	270	12	9.7	23x	28x	2.7	4.8	2.5	0.56x	1.1x	1.4	1.4	1.4	1.0x	1.0x
NT10K	Q_{desc}	Leaf parent	250	12	9.8	21x	25x	3.0	5.9	2.2	0.51x	1.4x	1.2	1.8	1.9	0.67x	0.65x
NT100K	Q_{desc}	Root	19M	150	140	130Kx	140Kx	260	92	77	2.8x	3.4x	4.3K	5.3K	5.3K	0.82x	0.81x
NT100K	Q_{desc}	Leaf parent H	2.5K	140	120	18x	20x	3.8	51	23	0.07x	0.17x	2.1	2.5	3.3	0.86x	0.65x
NT100K	Q_{desc}	Leaf parent	470	140	110	3.4x	4.1x	2.5	51	27	0.05x	0.09x	1.4	1.6	1.9	0.90x	0.77x
DT10	Q_{desc}	Root	9.5	1.4	1.4	7.0x	6.7x	2.9	1.5	1.7	2.0x	1.7x	1.2	2.1	2.2	0.54x	0.53x
DT10	Q_{desc}	Leaf parent H	9.3	1.4	1.4	6.8x	6.8x	2.8	1.5	1.6	1.8x	1.8x	1.7	1.9	1.9	0.91x	0.89x
DT10	Q_{desc}	Leaf parent	9.0	1.4	1.4	6.6x	6.7x	2.8	1.6	1.6	1.8x	1.7x	1.3	1.3	1.4	0.97x	0.97x
DT100	Q_{desc}	Root	20	1.5	1.5	14x	14x	3.9	1.5	1.8	2.6x	2.2x	12	13	14	0.86x	0.84x
DT100	Q_{desc}	Leaf parent H	9.3	1.4	1.4	6.5x	6.6x	3.0	1.5	1.7	2.0x	1.8x	1.7	1.8	1.8	0.95x	0.95x
DT100	Q_{desc}	Leaf parent	9.3	1.4	1.5	6.5x	6.4x	2.8	1.6	1.6	1.8x	1.7x	1.5	1.6	1.6	0.94x	0.92x
DT1K	Q_{desc}	Root	1.1K	2.8	2.8	390x	390x	11	1.7	1.8	6.6x	6.2x	40	47	49	0.84x	0.81x
DT1K	Q_{desc}	Leaf parent H	14	2.3	2.2	5.9x	6.2x	3.5	1.7	1.6	2.1x	2.2x	0.86	0.91	0.88	0.95x	0.99x
DT1K	Q_{desc}	Leaf parent	11	2.3	2.2	4.7x	4.9x	3.7	1.6	1.5	2.4x	2.4x	0.72	0.79	1.0	0.92x	0.69x
DT10K	Q_{desc}	Root	110K	16	15	6.7Kx	7.0Kx	32	7.1	8.8	4.5x	3.6x	420	490	510	0.85x	0.82x
DT10K	Q_{desc}	Leaf parent H	250	12	9.8	21x	25x	2.6	4.6	2.7	0.56x	0.98x	1.2	1.4	1.6	0.89x	0.77x
DT10K	Q_{desc}	Leaf parent	220	12	9.2	19x	24x	2.6	5.3	2.8	0.49x	0.92x	1.4	1.5	1.2	0.93x	1.2x
DT100K	Q_{desc}	Root	**>6 h**	140	140	**>150Kx**	**>150Kx**	130	93	77	1.5x	1.7x	4.3K	5.2K	5.4K	0.83x	0.80x
DT100K	Q_{desc}	Leaf parent H	800	140	120	5.6x	6.6x	3.4	51	27	0.07x	0.13x	1.3	1.8	1.8	0.74x	0.76x
DT100K	Q_{desc}	Leaf parent	470	140	120	3.4x	3.9x	2.3	51	13	0.04x	0.17x	1.6	1.6	1.6	0.99x	1.0x

Queries: Q_{desc} = All Descendants; Q_{child} = All Children; Q_{leaf} = All Leaves; $Q_{a\&d}$ = Check if Ancestor.
Parameters: Root = root of largest tree (also deepest when identical); Deep root = root of deepest tree.
Leaf parent = low-degree leaf parent; Leaf parent H = high-degree leaf parent; H/L = identical high/low-degree cases.
Methods: *B* = Baseline; *D* = Dewey; *P* = PrePost; S_D = *B* / *D*; S_P = *B* / *P*. Runtimes: median of five runs in ms.
Graphs: F = forest; NT = truebase; DT = ultratall; WT = ultrawide; K = 1,000 nodes.
SNB/C, SNB/P, SNB/T = SNB SF1 trees; SNB = full SNB SF1 graph.
Rounding: runtimes and speedups to about two significant digits; K = 1,000 and M = 1,000,000. Bounds round outward.
Timeouts: each >T runtime cell gives the user-confirmed limit T for that row and method (unit shown).
Speedup bounds use those limits in the same row/system; > is a lower bound, < an upper bound.

Speedup color: 

gray = slowdown; yellow = 1x; green = speedup; darker green = larger speedup

All Descendants (Query 01)

Graph	Query	Parameters	Apache AGE					Kuzu					Neo4j				
			B	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P
WT10	Q_{desc}	Root	9.4	1.4	1.4	6.8x	6.6x	3.0	1.6	1.5	1.9x	2.0x	2.0	2.2	2.2	0.93x	0.91x
WT10	Q_{desc}	Leaf parent H	9.3	1.4	1.3	6.6x	7.0x	3.1	1.6	1.5	1.9x	2.1x	1.7	1.9	2.0	0.89x	0.85x
WT10	Q_{desc}	Leaf parent	9.0	1.4	1.4	6.6x	6.6x	2.8	1.4	1.5	2.0x	1.9x	1.4	1.5	1.5	0.97x	0.96x
WT100	Q_{desc}	Root	26	2.2	1.8	12x	14x	3.5	1.5	1.5	2.3x	2.3x	12	13	14	0.88x	0.86x
WT100	Q_{desc}	Leaf parent H	10	1.4	1.4	7.1x	7.1x	2.7	1.7	1.4	1.6x	1.9x	2.5	2.7	2.8	0.93x	0.91x
WT100	Q_{desc}	Leaf parent	12	1.7	1.7	6.7x	6.8x	3.0	1.5	1.5	2.0x	2.0x	1.7	2.1	1.8	0.82x	0.95x
WT1K	Q_{desc}	Root	1.2K	2.6	2.7	440x	420x	8.7	1.6	2.2	5.3x	3.9x	59	45	47	1.3x	1.3x
WT1K	Q_{desc}	Leaf parent H	20	2.3	2.3	8.8x	8.7x	3.3	1.9	1.8	1.7x	1.8x	1.6	2.0	2.6	0.77x	0.60x
WT1K	Q_{desc}	Leaf parent	22	2.3	1.7	9.7x	13x	3.5	1.6	1.7	2.2x	2.0x	2.2	2.3	2.5	0.93x	0.88x
WT10K	Q_{desc}	Root	110K	8.6	11	13Kx	10Kx	25	10	8.6	2.4x	2.9x	410	490	500	0.84x	0.81x
WT10K	Q_{desc}	Leaf parent H	330	12	10	29x	33x	2.4	5.2	3.0	0.47x	0.81x	1.7	2.2	2.8	0.78x	0.61x
WT10K	Q_{desc}	Leaf parent	110	5.7	1.9	20x	60x	3.0	4.6	2.9	0.65x	1.0x	2.4	2.6	2.6	0.92x	0.93x
WT100K	Q_{desc}	Root	24M	76	74	310Kx	320Kx	120	92	76	1.3x	1.6x	4.3K	5.1K	5.3K	0.84x	0.82x
WT100K	Q_{desc}	Leaf parent H	3.0K	140	130	21x	23x	3.5	50	26	0.07x	0.13x	2.3	2.6	2.6	0.89x	0.90x
WT100K	Q_{desc}	Leaf parent	580	37	1.9	15x	300x	2.2	51	27	0.04x	0.08x	1.2	1.3	1.3	0.93x	0.92x
SNB/C	Q_{desc}	Root	92K	680	2.0	140x	47Kx	3.6	420	390	0.01x	0.01x	4.7	4.5	4.7	1.0x	1.0x
SNB/C	Q_{desc}	Deep root	82K	2.6K	2.1K	31x	39x	4.1	390	400	0.01x	0.01x	4.1	4.4	4.1	0.92x	1.0x
SNB/C	Q_{desc}	Leaf parent H	69K	2.6K	2.4K	26x	28x	3.6	410	370	0.01x	0.01x	3.5	3.8	4.0	0.93x	0.87x
SNB/C	Q_{desc}	Leaf parent	22K	720	2.0	31x	11Kx	3.4	440	390	0.01x	0.01x	1.6	1.6	1.6	1.0x	0.99x
SNB/P	Q_{desc}	Root	17K	2.7	2.6	6.2Kx	6.6Kx	6.9	2.5	2.3	2.8x	3.1x	46	41	43	1.1x	1.1x
SNB/P	Q_{desc}	Leaf parent H	15K	3.0	2.9	5.1Kx	5.1Kx	3.7	2.0	1.8	1.9x	2.0x	11	13	14	0.85x	0.79x
SNB/P	Q_{desc}	Leaf parent	15K	1.6	1.9	9.0Kx	7.8Kx	3.8	2.1	1.9	1.8x	2.0x	1.1	1.3	1.5	0.87x	0.74x
SNB/T	Q_{desc}	Root	14K	1.9	1.9	7.7Kx	7.6Kx	3.3	1.7	1.6	2.0x	2.1x	9.7	11	11	0.89x	0.85x
SNB/T	Q_{desc}	Leaf parent H	15K	1.5	1.5	9.9Kx	10Kx	3.1	1.6	1.7	1.9x	1.8x	8.6	8.9	9.1	0.96x	0.94x
SNB/T	Q_{desc}	Leaf parent	14K	1.6	1.7	9.0Kx	8.6Kx	2.9	1.6	1.5	1.9x	2.0x	1.5	1.5	1.5	1.0x	1.0x

Queries: Q_{desc} = All Descendants; Q_{child} = All Children; Q_{leaf} = All Leaves; $Q_{a\&d}$ = Check if Ancestor.
Parameters: Root = root of largest tree (also deepest when identical); Deep root = root of deepest tree.
Leaf parent = low-degree leaf parent; Leaf parent H = high-degree leaf parent; H/L = identical high/low-degree cases.
Methods: *B* = Baseline; *D* = Dewey; *P* = PrePost; S_D = *B* / *D*; S_P = *B* / *P*. Runtimes: median of five runs in ms.
Graphs: F = forest; NT = truebase; DT = ultratall; WT = ultrawide; K = 1,000 nodes.
SNB/C, SNB/P, SNB/T = SNB SF1 trees; SNB = full SNB SF1 graph.
Rounding: runtimes and speedups to about two significant digits; K = 1,000 and M = 1,000,000. Bounds round outward.

Speedup color: 

gray = slowdown; yellow = 1x; green = speedup; darker green = larger speedup

All Children (Query 02)

Graph	Query	Parameters	Apache AGE					Kuzu					Neo4j				
			B	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P
F40	Q_{child}	High degree	2.0	1.6	1.5	1.2x	1.3x	1.7	1.8	1.8	0.91x	0.93x	1.7	1.6	1.6	1.1x	1.0x
F40	Q_{child}	Low degree	1.9	1.6	1.5	1.2x	1.3x	2.0	1.7	1.9	1.2x	1.0x	1.2	1.1	1.1	1.1x	1.1x
F1K	Q_{child}	High degree	2.4	2.5	2.3	0.97x	1.0x	1.6	2.5	2.5	0.63x	0.64x	1.9	1.9	1.6	0.99x	1.2x
F1K	Q_{child}	Low degree	2.3	2.4	2.2	0.95x	1.0x	1.5	2.3	1.4	0.65x	1.1x	1.1	1.1	1.1	0.97x	0.96x
NT10	Q_{child}	High degree	1.9	1.6	1.3	1.2x	1.5x	1.7	2.0	1.9	0.82x	0.89x	1.6	1.7	1.7	0.94x	0.94x
NT10	Q_{child}	Low degree	1.9	1.6	1.4	1.2x	1.3x	1.6	1.8	1.3	0.94x	1.3x	1.3	1.1	1.1	1.2x	1.2x
NT100	Q_{child}	High degree	2.0	1.5	1.5	1.3x	1.3x	1.8	1.7	1.7	1.1x	1.1x	2.0	2.1	2.2	0.97x	0.91x
NT100	Q_{child}	Low degree	1.9	1.6	1.5	1.2x	1.3x	1.7	1.7	1.8	1.0x	0.95x	1.3	1.2	1.4	1.1x	0.89x
NT1K	Q_{child}	High degree	2.4	2.4	2.3	1.0x	1.0x	1.7	1.9	2.0	0.89x	0.88x	1.2	1.1	1.1	1.1x	1.1x
NT1K	Q_{child}	Low degree	2.2	2.3	2.2	0.97x	1.0x	1.7	1.9	1.8	0.91x	0.96x	0.71	0.73	0.68	0.97x	1.0x
NT10K	Q_{child}	High degree	6.3	11	10	0.60x	0.61x	2.2	4.4	3.0	0.49x	0.72x	1.9	1.9	1.9	0.98x	1.0x
NT10K	Q_{child}	Low degree	5.4	10	10	0.52x	0.53x	2.2	5.1	2.4	0.43x	0.91x	1.1	1.2	1.1	0.94x	0.97x
NT100K	Q_{child}	High degree	57	110	100	0.53x	0.55x	2.2	39	27	0.06x	0.08x	3.5	3.9	4.1	0.92x	0.86x
NT100K	Q_{child}	Low degree	56	110	100	0.53x	0.54x	1.9	39	27	0.05x	0.07x	1.3	1.4	1.3	0.93x	0.98x
DT10	Q_{child}	High degree	2.0	1.6	1.5	1.3x	1.3x	1.8	1.4	1.5	1.3x	1.2x	1.5	1.7	1.6	0.89x	0.95x
DT10	Q_{child}	Low degree	1.9	1.6	1.5	1.2x	1.3x	1.6	1.8	1.7	0.90x	0.92x	1.1	1.1	1.1	0.97x	0.97x
DT100	Q_{child}	High degree	2.0	1.6	1.6	1.2x	1.3x	1.4	1.4	1.7	0.99x	0.82x	1.7	1.6	1.5	1.1x	1.1x
DT100	Q_{child}	Low degree	1.9	1.6	1.5	1.2x	1.3x	1.9	1.6	1.8	1.2x	1.1x	1.0	1.1	1.1	0.94x	0.94x
DT1K	Q_{child}	High degree	2.4	2.4	2.3	1.0x	1.1x	1.1	1.8	1.9	0.64x	0.60x	1.5	1.5	1.4	0.98x	1.1x
DT1K	Q_{child}	Low degree	2.3	2.4	2.3	0.94x	1.0x	1.8	1.8	1.8	1.0x	1.0x	0.95	1.0	1.0	0.93x	0.92x
DT10K	Q_{child}	High degree	6.3	11	10	0.60x	0.61x	1.7	4.9	2.8	0.34x	0.59x	1.5	1.5	1.5	1.0x	1.0x
DT10K	Q_{child}	Low degree	5.4	10	10	0.52x	0.53x	1.8	4.3	2.6	0.41x	0.67x	1.0	0.99	0.98	1.0x	1.0x
DT100K	Q_{child}	High degree	65	110	100	0.62x	0.66x	1.7	39	20	0.04x	0.08x	1.8	1.7	1.7	1.0x	1.1x
DT100K	Q_{child}	Low degree	56	110	100	0.53x	0.54x	1.8	39	20	0.05x	0.09x	1.2	1.3	1.2	0.90x	1.0x

Queries: Q_{desc} = All Descendants; Q_{child} = All Children; Q_{leaf} = All Leaves; $Q_{a\&d}$ = Check if Ancestor.
Parameters: High degree = highest-degree node; Low degree = lowest-degree node.
Methods: *B* = Baseline; *D* = Dewey; *P* = PrePost; S_D = *B* / *D*; S_P = *B* / *P*. Runtimes: median of five runs in ms.
Graphs: F = forest; NT = truebase; DT = ultratall; WT = ultrawide; K = 1,000 nodes.
SNB/C, SNB/P, SNB/T = SNB SF1 trees; SNB = full SNB SF1 graph.
Rounding: runtimes and speedups to about two significant digits; K = 1,000 and M = 1,000,000. Bounds round outward.

Speedup color: 

gray = slowdown; yellow = 1x; green = speedup; darker green = larger speedup

All Children (Query 02)

Graph	Query	Parameters	Apache AGE					Kuzu					Neo4j				
			B	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P
WT10	Q_{child}	High degree	2.0	1.6	1.5	1.3x	1.4x	2.2	1.8	1.7	1.2x	1.3x	1.7	1.7	1.7	0.97x	0.99x
WT10	Q_{child}	Low degree	1.8	1.6	1.4	1.2x	1.3x	1.7	1.6	1.3	1.0x	1.3x	1.1	1.1	1.2	0.99x	0.95x
WT100	Q_{child}	High degree	2.0	1.7	1.6	1.2x	1.3x	1.6	1.7	1.8	0.92x	0.87x	2.0	2.1	1.8	0.91x	1.1x
WT100	Q_{child}	Low degree	2.0	1.7	1.5	1.2x	1.3x	1.6	1.4	1.5	1.1x	1.1x	1.1	1.2	1.2	0.98x	0.96x
WT1K	Q_{child}	High degree	2.5	2.4	2.4	1.0x	1.0x	1.6	1.8	1.8	0.89x	0.86x	2.3	2.5	2.4	0.95x	0.99x
WT1K	Q_{child}	Low degree	2.3	2.3	2.3	1.0x	0.99x	1.6	1.7	1.6	0.93x	1.0x	1.1	1.2	1.2	0.95x	0.92x
WT10K	Q_{child}	High degree	6.3	11	10	0.59x	0.60x	1.5	5.0	2.9	0.29x	0.51x	2.6	2.9	2.7	0.88x	0.96x
WT10K	Q_{child}	Low degree	5.4	10	10	0.52x	0.52x	1.5	4.2	2.6	0.35x	0.58x	1.2	1.2	1.2	0.98x	0.98x
WT100K	Q_{child}	High degree	63	100	100	0.62x	0.63x	1.6	39	27	0.04x	0.06x	1.5	1.8	1.8	0.87x	0.84x
WT100K	Q_{child}	Low degree	54	100	99	0.52x	0.55x	2.0	39	22	0.05x	0.09x	0.74	0.68	0.69	1.1x	1.1x
SNB/C	Q_{child}	High degree	1.4K	3.6K	3.5K	0.40x	0.41x	2.8	370	390	0.01x	0.01x	3.4	3.8	3.8	0.90x	0.89x
SNB/C	Q_{child}	Low degree	1.2K	3.5K	3.2K	0.33x	0.36x	1.8	360	360	0.01x	0.01x	1.2	1.3	0.99	0.92x	1.2x
SNB/P	Q_{child}	High degree	3.5	3.3	3.1	1.1x	1.2x	2.4	2.3	2.1	1.0x	1.1x	24	26	26	0.95x	0.93x
SNB/P	Q_{child}	Low degree	2.4	2.9	2.5	0.82x	0.94x	1.5	2.4	2.4	0.62x	0.63x	1.2	1.3	1.3	0.91x	0.92x
SNB/T	Q_{child}	High degree	2.1	1.7	1.6	1.3x	1.4x	1.7	1.6	1.8	1.0x	0.94x	3.7	4.2	4.1	0.89x	0.90x
SNB/T	Q_{child}	Low degree	1.9	1.6	1.5	1.2x	1.3x	1.7	1.5	1.7	1.1x	1.0x	1.2	1.1	1.3	1.1x	0.94x

Queries: Q_{desc} = All Descendants; Q_{child} = All Children; Q_{leaf} = All Leaves; $Q_{a\&d}$ = Check if Ancestor.
Parameters: High degree = highest-degree node; Low degree = lowest-degree node.
Methods: *B* = Baseline; *D* = Dewey; *P* = PrePost; S_D = *B* / *D*; S_P = *B* / *P*. Runtimes: median of five runs in ms.
Graphs: F = forest; NT = truebase; DT = ultratall; WT = ultrawide; K = 1,000 nodes.
SNB/C, SNB/P, SNB/T = SNB SF1 trees; SNB = full SNB SF1 graph.
Rounding: runtimes and speedups to about two significant digits; K = 1,000 and M = 1,000,000. Bounds round outward.

Speedup color: 

gray = slowdown; yellow = 1x; green = speedup; darker green = larger speedup

All Leaves (Query 05)

Graph	Query	Parameters	Apache AGE					Kuzu					Neo4j				
			B	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P
F40	Q_{leaf}	Root	9.6	1.6	2.1	6.1x	4.7x	5.3	3.1	1.7	1.7x	3.2x	1.9	1.9	1.7	1.0x	1.1x
F40	Q_{leaf}	Leaf parent H	9.9	1.5	2.0	6.5x	4.8x	4.6	2.7	2.0	1.7x	2.3x	1.4	1.5	1.5	0.94x	0.94x
F40	Q_{leaf}	Leaf parent	9.6	1.5	2.0	6.5x	4.7x	4.2	3.2	1.9	1.3x	2.2x	1.4	1.5	1.7	0.96x	0.84x
F1K	Q_{leaf}	Root	310	19	2.4	16x	130x	8.0	3.0	2.2	2.7x	3.6x	11	26	26	0.43x	0.43x
F1K	Q_{leaf}	Leaf parent H	96	2.4	2.3	40x	43x	5.6	3.5	1.9	1.6x	2.9x	2.3	2.7	2.1	0.86x	1.1x
F1K	Q_{leaf}	Leaf parent	95	2.4	2.3	40x	42x	6.4	2.6	1.8	2.4x	3.6x	1.9	1.9	1.7	0.96x	1.1x
NT10	Q_{leaf}	Root	9.7	1.5	1.4	6.6x	6.9x	3.8	2.8	1.9	1.4x	2.0x	1.8	1.9	1.8	0.96x	0.98x
NT10	Q_{leaf}	Leaf parent H/L	9.0	1.4	1.4	6.3x	6.3x	4.6	2.9	1.9	1.6x	2.4x	1.6	1.7	1.5	0.93x	1.1x
NT100	Q_{leaf}	Root	18	2.8	6.3	6.4x	2.9x	6.1	3.1	1.6	2.0x	3.8x	9.4	11	9.8	0.84x	0.96x
NT100	Q_{leaf}	Leaf parent H	11	1.5	6.1	7.4x	1.8x	5.0	2.9	1.8	1.7x	2.8x	2.2	2.4	2.1	0.91x	1.1x
NT100	Q_{leaf}	Leaf parent	11	1.5	6.1	7.2x	1.8x	5.4	2.9	1.6	1.8x	3.3x	1.8	1.8	1.6	1.0x	1.1x
NT1K	Q_{leaf}	Root	950	130	2.8	7.5x	340x	13	3.2	2.2	3.9x	5.6x	31	35	35	0.88x	0.87x
NT1K	Q_{leaf}	Leaf parent H	160	2.3	2.3	71x	70x	4.5	3.6	2.0	1.2x	2.2x	1.4	1.7	1.5	0.86x	0.96x
NT1K	Q_{leaf}	Leaf parent	160	2.5	2.3	65x	69x	5.2	3.3	1.7	1.5x	3.0x	1.2	1.2	1.1	1.0x	1.1x
NT10K	Q_{leaf}	Root	93K	13K	14	7.4x	6.4Kx	49	20	7.6	2.4x	6.4x	290	340	350	0.87x	0.84x
NT10K	Q_{leaf}	Leaf parent H	12K	12	11	1.0Kx	1.1Kx	5.7	18	2.4	0.32x	2.3x	1.4	1.9	2.1	0.75x	0.66x
NT10K	Q_{leaf}	Leaf parent	12K	12	11	1.0Kx	1.1Kx	5.1	16	2.3	0.32x	2.2x	1.8	1.9	1.9	0.92x	0.96x
NT100K	Q_{leaf}	Root	19M	1.2M	110	16x	170Kx	110	200	74	0.53x	1.5x	4.1K	4.9K	5.1K	0.83x	0.81x
NT100K	Q_{leaf}	Leaf parent H	1.2M	340	110	3.6Kx	11Kx	5.7	160	21	0.04x	0.27x	2.4	2.5	2.4	0.97x	1.0x
NT100K	Q_{leaf}	Leaf parent	1.2M	340	110	3.6Kx	11Kx	4.6	170	20	0.03x	0.23x	1.3	1.2	1.0	1.0x	1.2x
DT10	Q_{leaf}	Root	9.4	1.5	1.4	6.2x	6.5x	3.9	2.7	1.7	1.5x	2.3x	1.8	1.8	2.0	0.97x	0.90x
DT10	Q_{leaf}	Leaf parent H	9.6	1.4	1.4	6.8x	6.7x	3.6	2.6	1.6	1.4x	2.2x	2.0	1.9	1.9	1.0x	1.0x
DT10	Q_{leaf}	Leaf parent	9.7	1.3	1.5	7.4x	6.6x	3.9	2.6	1.7	1.5x	2.3x	1.5	1.6	1.5	0.95x	1.0x
DT100	Q_{leaf}	Root	16	2.8	5.4	5.7x	3.0x	4.5	2.8	1.8	1.6x	2.5x	6.8	9.0	8.2	0.75x	0.83x
DT100	Q_{leaf}	Leaf parent H	11	1.5	5.2	7.3x	2.1x	4.1	2.5	1.7	1.6x	2.5x	1.8	1.9	1.8	0.94x	1.0x
DT100	Q_{leaf}	Leaf parent	11	1.5	5.3	7.1x	2.0x	4.1	2.9	1.8	1.4x	2.3x	1.6	1.6	1.6	0.96x	1.0x
DT1K	Q_{leaf}	Root	650	120	2.6	5.4x	240x	13	2.8	2.0	4.7x	6.5x	38	34	30	1.1x	1.3x
DT1K	Q_{leaf}	Leaf parent H	97	2.4	2.3	40x	43x	4.8	3.2	2.0	1.5x	2.4x	1.3	1.9	1.9	0.72x	0.72x
DT1K	Q_{leaf}	Leaf parent	96	2.4	2.3	41x	42x	5.1	2.9	1.8	1.8x	2.8x	1.6	1.7	1.6	0.99x	1.0x
DT10K	Q_{leaf}	Root	60K	11K	13	5.5x	4.5Kx	30	11	5.3	2.9x	5.7x	200	230	230	0.87x	0.86x
DT10K	Q_{leaf}	Leaf parent H	6.4K	12	11	550x	610x	3.8	13	1.8	0.29x	2.1x	1.3	1.6	1.8	0.81x	0.70x
DT10K	Q_{leaf}	Leaf parent	6.4K	12	11	550x	610x	5.2	18	1.9	0.29x	2.7x	1.6	1.6	1.6	1.0x	1.0x
DT100K	Q_{leaf}	Root	12M	900K	97	13x	120Kx	140	160	42	0.87x	3.2x	2.1K	2.6K	2.6K	0.83x	0.81x
DT100K	Q_{leaf}	Leaf parent H	620K	340	110	1.8Kx	5.6Kx	5.9	130	12	0.05x	0.51x	1.3	1.4	1.3	0.88x	0.98x
DT100K	Q_{leaf}	Leaf parent	620K	340	110	1.8Kx	5.7Kx	4.5	120	12	0.04x	0.36x	1.1	1.2	1.0	0.94x	1.1x

Queries: Q_{desc} = All Descendants; Q_{child} = All Children; Q_{leaf} = All Leaves; $Q_{a\&d}$ = Check if Ancestor.
Parameters: Root = root of largest tree (also deepest when identical); Deep root = root of deepest tree.
Leaf parent = low-degree leaf parent; Leaf parent H = high-degree leaf parent; H/L = identical high/low-degree cases.
Methods: *B* = Baseline; *D* = Dewey; *P* = PrePost; $S_D = B / D$; $S_P = B / P$. Runtimes: median of five runs in ms.
Graphs: F = forest; NT = truebase; DT = ultratall; WT = ultrawide; K = 1,000 nodes.
SNB/C, SNB/P, SNB/T = SNB SF1 trees; SNB = full SNB SF1 graph.
Rounding: runtimes and speedups to about two significant digits; K = 1,000 and M = 1,000,000. Bounds round outward.

Speedup color: 

gray = slowdown; yellow = 1x; green = speedup; darker green = larger speedup

All Leaves (Query 05)

Graph	Query	Parameters	Apache AGE					Kuzu					Neo4j				
			B	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P
WT10	Q_{leaf}	Root	9.6	1.5	1.5	6.4x	6.5x	4.1	2.6	1.6	1.5x	2.6x	1.9	1.9	1.9	0.96x	1.0x
WT10	Q_{leaf}	Leaf parent H	9.2	1.4	1.4	6.5x	6.4x	4.3	2.9	1.9	1.5x	2.3x	1.6	1.9	1.8	0.88x	0.90x
WT10	Q_{leaf}	Leaf parent	9.6	1.4	1.4	6.7x	6.7x	3.7	2.5	1.5	1.5x	2.5x	1.5	1.6	1.5	0.94x	0.99x
WT100	Q_{leaf}	Root	20	2.9	7.6	7.1x	2.7x	5.0	3.0	1.9	1.7x	2.7x	7.3	11	12	0.65x	0.61x
WT100	Q_{leaf}	Leaf parent H	12	1.6	7.5	7.6x	1.6x	3.7	2.7	1.9	1.4x	1.9x	2.8	3.0	3.0	0.91x	0.94x
WT100	Q_{leaf}	Leaf parent	11	1.6	7.3	7.1x	1.5x	3.8	2.7	1.9	1.4x	2.0x	2.1	2.2	2.1	0.94x	0.98x
WT1K	Q_{leaf}	Root	1.1K	130	2.8	8.4x	410x	9.4	3.2	1.9	3.0x	4.9x	45	40	42	1.1x	1.1x
WT1K	Q_{leaf}	Leaf parent H	170	2.4	2.3	70x	73x	4.7	3.5	1.9	1.4x	2.4x	1.7	1.8	1.7	0.94x	0.98x
WT1K	Q_{leaf}	Leaf parent	170	2.4	2.3	69x	72x	4.8	3.4	1.6	1.4x	3.0x	1.5	1.6	1.6	0.94x	0.98x
WT10K	Q_{leaf}	Root	110K	14K	15	8.1x	7.2Kx	30	25	8.1	1.2x	3.7x	380	400	460	0.94x	0.82x
WT10K	Q_{leaf}	Leaf parent H	13K	12	11	1.1Kx	1.1Kx	3.7	21	2.5	0.18x	1.5x	2.1	2.6	3.2	0.78x	0.64x
WT10K	Q_{leaf}	Leaf parent	13K	12	11	1.1Kx	1.1Kx	4.5	21	2.5	0.21x	1.8x	2.8	2.7	2.2	1.0x	1.3x
WT100K	Q_{leaf}	Root	18M	1.1M	110	17x	160Kx	110	200	73	0.58x	1.6x	4.0K	4.7K	4.9K	0.84x	0.81x
WT100K	Q_{leaf}	Leaf parent H	1.2M	340	110	3.6Kx	12Kx	5.4	160	26	0.03x	0.21x	2.3	3.0	4.3	0.78x	0.54x
WT100K	Q_{leaf}	Leaf parent	1.2M	340	110	3.6Kx	11Kx	4.3	170	21	0.03x	0.21x	1.8	2.0	1.9	0.90x	0.94x
SNB/C	Q_{leaf}	Root	**>6 h**	2.7K	2.3K	**>8.0Kx**	**>9.2Kx**	6.0	550	330	0.01x	0.02x	2.0	3.1	2.9	0.66x	0.70x
SNB/C	Q_{leaf}	Deep root	**>6 h**	2.7K	2.3K	**>7.9Kx**	**>9.2Kx**	5.9	570	290	0.01x	0.02x	2.7	2.9	3.1	0.91x	0.86x
SNB/C	Q_{leaf}	Leaf parent H	**>6 h**	2.7K	2.3K	**>8.0Kx**	**>9.2Kx**	5.2	590	340	0.01x	0.01x	3.6	3.9	3.9	0.93x	0.93x
SNB/C	Q_{leaf}	Leaf parent	**>6 h**	2.7K	2.3K	**>7.9Kx**	**>9.2Kx**	5.9	580	340	0.01x	0.02x	1.6	1.7	1.6	0.98x	1.0x
SNB/P	Q_{leaf}	Root	17K	81	3.2	210x	5.3Kx	9.6	4.3	2.5	2.2x	3.8x	46	40	41	1.1x	1.1x
SNB/P	Q_{leaf}	Leaf parent H	15K	8.8	3.0	1.8Kx	5.2Kx	5.1	4.0	2.1	1.3x	2.4x	12	14	15	0.81x	0.76x
SNB/P	Q_{leaf}	Leaf parent	15K	2.9	2.8	5.1Kx	5.3Kx	5.0	4.2	2.3	1.2x	2.2x	1.2	1.5	1.6	0.80x	0.80x
SNB/T	Q_{leaf}	Root	15K	2.3	4.4	6.4Kx	3.3Kx	4.6	2.5	1.8	1.9x	2.6x	10	9.8	9.5	1.0x	1.1x
SNB/T	Q_{leaf}	Leaf parent H	15K	2.0	4.3	7.3Kx	3.4Kx	4.4	2.6	1.7	1.7x	2.6x	8.2	6.1	8.6	1.3x	0.95x
SNB/T	Q_{leaf}	Leaf parent	14K	1.5	4.2	9.5Kx	3.4Kx	3.4	2.4	1.7	1.4x	2.0x	1.7	1.5	1.5	1.1x	1.1x

Queries: Q_{desc} = All Descendants; Q_{child} = All Children; Q_{leaf} = All Leaves; $Q_{a\&d}$ = Check if Ancestor.
Parameters: Root = root of largest tree (also deepest when identical); Deep root = root of deepest tree.
Leaf parent = low-degree leaf parent; Leaf parent H = high-degree leaf parent; H/L = identical high/low-degree cases.
Methods: *B* = Baseline; *D* = Dewey; *P* = PrePost; S_D = *B* / *D*; S_P = *B* / *P*. Runtimes: median of five runs in ms.
Graphs: F = forest; NT = truebase; DT = ultratall; WT = ultrawide; K = 1,000 nodes.
SNB/C, SNB/P, SNB/T = SNB SF1 trees; SNB = full SNB SF1 graph.
Rounding: runtimes and speedups to about two significant digits; K = 1,000 and M = 1,000,000. Bounds round outward.
Timeouts: each >T runtime cell gives the user-confirmed limit T for that row and method (unit shown).
Speedup bounds use those limits in the same row/system; > is a lower bound, < an upper bound.

Speedup color: 

gray = slowdown; yellow = 1x; green = speedup; darker green = larger speedup

Check if Ancestor (Query 11)

Graph	Query	Parameters	Apache AGE					Kuzu					Neo4j				
			B	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P
F40	$Q_{a\&d}$	Root–leaf	10	1.4	1.3	7.3x	7.9x	22	0.90	0.94	25x	24x	1.5	1.5	1.3	1.0x	1.1x
F40	$Q_{a\&d}$	Deep parent–leaf	10	1.1	1.2	9.2x	8.1x	20	0.90	0.93	22x	21x	1.3	1.3	1.5	0.98x	0.87x
F40	$Q_{a\&d}$	Siblings	9.9	1.1	1.1	9.0x	9.1x	20	0.97	0.91	21x	22x	1.5	1.3	1.3	1.2x	1.1x
F40	$Q_{a\&d}$	Distant leaves	9.3	1.4	1.1	6.7x	8.5x	21	0.90	0.94	24x	23x	1.5	1.4	1.4	1.1x	1.0x
F1K	$Q_{a\&d}$	Root–leaf	12	1.4	1.5	8.8x	8.4x	19	1.0	0.91	19x	21x	1.7	1.1	1.6	1.5x	1.1x
F1K	$Q_{a\&d}$	Deep parent–leaf	12	1.6	1.5	7.5x	8.4x	22	1.0	0.92	22x	24x	1.8	1.2	1.6	1.6x	1.2x
F1K	$Q_{a\&d}$	Siblings	11	2.6	2.6	4.2x	4.2x	17	1.1	0.88	16x	19x	1.6	1.3	1.6	1.2x	0.96x
F1K	$Q_{a\&d}$	Distant leaves	10	2.5	2.6	4.1x	4.0x	22	0.96	1.0	23x	21x	1.7	1.3	1.6	1.3x	1.0x
NT10	$Q_{a\&d}$	Root–leaf	9.8	1.2	1.1	8.0x	9.1x	19	0.94	0.94	20x	20x	1.6	1.3	1.2	1.2x	1.3x
NT10	$Q_{a\&d}$	Deep parent–leaf	9.8	1.1	1.2	8.9x	7.9x	19	1.0	0.95	19x	20x	1.3	1.2	1.1	1.1x	1.2x
NT10	$Q_{a\&d}$	Siblings	9.3	1.3	0.98	7.3x	9.5x	21	1.0	0.98	21x	21x	1.5	1.4	1.7	1.0x	0.87x
NT10	$Q_{a\&d}$	Distant leaves	9.4	1.0	0.99	9.4x	9.4x	20	0.89	0.94	23x	22x	1.4	1.2	1.2	1.2x	1.1x
NT100	$Q_{a\&d}$	Root–leaf	10	1.3	1.3	7.6x	7.6x	21	0.95	0.91	22x	23x	2.0	1.4	0.75	1.5x	2.7x
NT100	$Q_{a\&d}$	Deep parent–leaf	10	1.3	1.3	7.9x	7.7x	21	0.93	0.98	22x	21x	1.4	1.3	1.5	1.1x	0.96x
NT100	$Q_{a\&d}$	Siblings	9.3	1.2	1.1	8.0x	8.2x	19	0.95	0.93	20x	20x	1.7	1.3	1.2	1.3x	1.5x
NT100	$Q_{a\&d}$	Distant leaves	9.5	1.2	1.1	8.3x	8.5x	22	0.92	0.94	24x	23x	1.2	0.89	0.88	1.3x	1.3x
NT1K	$Q_{a\&d}$	Root–leaf	11	1.5	1.4	7.4x	7.9x	20	0.96	0.89	20x	22x	1.2	0.82	0.71	1.5x	1.7x
NT1K	$Q_{a\&d}$	Deep parent–leaf	11	1.4	1.3	7.6x	8.1x	19	0.83	0.89	23x	22x	0.86	1.1	1.1	0.78x	0.79x
NT1K	$Q_{a\&d}$	Siblings	11	2.6	2.8	4.2x	3.9x	22	0.95	0.86	24x	26x	1.4	1.1	1.2	1.3x	1.2x
NT1K	$Q_{a\&d}$	Distant leaves	11	2.5	2.6	4.2x	4.1x	21	0.89	0.95	23x	22x	1.4	1.1	0.93	1.4x	1.5x
NT10K	$Q_{a\&d}$	Root–leaf	18	1.6	1.4	12x	13x	22	0.88	0.93	26x	24x	1.6	1.3	1.4	1.2x	1.2x
NT10K	$Q_{a\&d}$	Deep parent–leaf	18	1.5	1.6	12x	11x	22	0.86	0.87	26x	26x	1.3	1.1	1.2	1.2x	1.2x
NT10K	$Q_{a\&d}$	Siblings	21	16	16	1.3x	1.3x	21	0.89	0.90	24x	23x	1.4	1.2	1.3	1.1x	1.1x
NT10K	$Q_{a\&d}$	Distant leaves	21	16	16	1.3x	1.3x	25	0.84	0.95	30x	26x	1.6	1.2	1.2	1.3x	1.3x
NT100K	$Q_{a\&d}$	Root–leaf	79	1.6	1.6	49x	48x	20	1.1	0.86	18x	23x	1.1	0.95	0.84	1.2x	1.4x
NT100K	$Q_{a\&d}$	Deep parent–leaf	79	1.6	1.6	49x	49x	22	0.95	0.91	23x	24x	0.93	0.80	0.83	1.2x	1.1x
NT100K	$Q_{a\&d}$	Siblings	130	160	160	0.80x	0.80x	18	1.0	0.89	18x	21x	0.84	0.84	0.81	1.0x	1.0x
NT100K	$Q_{a\&d}$	Distant leaves	130	160	160	0.79x	0.79x	20	1.0	0.94	19x	21x	0.90	0.79	0.85	1.1x	1.1x
DT10	$Q_{a\&d}$	Root–leaf	9.8	1.3	1.3	7.3x	7.4x	17	0.89	0.96	20x	18x	1.4	1.2	1.1	1.2x	1.3x
DT10	$Q_{a\&d}$	Deep parent–leaf	9.7	1.1	1.3	9.0x	7.6x	18	0.87	1.0	21x	18x	1.3	1.2	1.2	1.1x	1.1x
DT10	$Q_{a\&d}$	Siblings	9.4	1.1	1.0	8.8x	9.2x	19	0.85	0.97	22x	20x	1.2	1.2	1.2	1.0x	1.0x
DT10	$Q_{a\&d}$	Distant leaves	9.4	1.0	0.98	9.2x	9.6x	18	0.87	0.92	21x	20x	1.3	1.2	1.2	1.1x	1.1x
DT100	$Q_{a\&d}$	Root–leaf	9.7	1.1	1.3	8.5x	7.3x	20	0.84	0.90	24x	23x	1.5	1.1	1.1	1.3x	1.4x
DT100	$Q_{a\&d}$	Deep parent–leaf	10	1.4	1.3	7.5x	7.6x	22	0.94	0.93	23x	23x	1.2	1.1	1.2	1.1x	1.0x
DT100	$Q_{a\&d}$	Siblings	9.5	1.2	1.1	8.0x	8.3x	19	1.0	0.89	19x	22x	1.3	1.2	1.2	1.1x	1.0x
DT100	$Q_{a\&d}$	Distant leaves	9.6	1.2	1.1	8.1x	8.5x	19	0.94	0.92	20x	20x	1.4	1.3	0.67	1.1x	2.1x
DT1K	$Q_{a\&d}$	Root–leaf	11	1.4	1.3	7.6x	8.6x	20	0.85	0.98	24x	21x	1.8	1.5	1.2	1.3x	1.5x
DT1K	$Q_{a\&d}$	Deep parent–leaf	11	1.2	1.2	8.7x	9.0x	20	0.94	0.94	21x	21x	1.3	1.4	1.2	0.97x	1.1x
DT1K	$Q_{a\&d}$	Siblings	11	2.7	2.5	4.2x	4.4x	18	0.90	0.92	19x	19x	1.4	1.3	1.3	1.1x	1.0x
DT1K	$Q_{a\&d}$	Distant leaves	11	2.6	2.5	4.1x	4.2x	20	0.97	0.94	21x	21x	1.6	1.3	1.3	1.2x	1.2x
DT10K	$Q_{a\&d}$	Root–leaf	19	1.6	1.6	12x	12x	22	0.91	1.0	24x	21x	1.6	1.2	1.1	1.4x	1.5x
DT10K	$Q_{a\&d}$	Deep parent–leaf	18	1.5	1.6	12x	11x	20	0.94	0.91	21x	22x	1.3	1.1	1.1	1.2x	1.2x
DT10K	$Q_{a\&d}$	Siblings	21	17	17	1.3x	1.3x	20	0.87	1.0	23x	20x	1.4	1.2	1.2	1.1x	1.1x
DT10K	$Q_{a\&d}$	Distant leaves	21	17	17	1.3x	1.3x	20	0.98	0.95	21x	21x	1.6	1.2	1.2	1.3x	1.3x
DT100K	$Q_{a\&d}$	Root–leaf	80	1.6	1.6	49x	50x	20	1.0	1.0	19x	19x	1.1	1.0	0.85	1.1x	1.3x
DT100K	$Q_{a\&d}$	Deep parent–leaf	81	1.4	1.3	60x	60x	21	1.0	0.93	21x	23x	0.85	0.82	1.1	1.0x	0.75x
DT100K	$Q_{a\&d}$	Siblings	130	160	160	0.82x	0.84x	20	0.97	0.97	21x	21x	1.2	1.2	1.3	0.94x	0.92x
DT100K	$Q_{a\&d}$	Distant leaves	130	160	160	0.80x	0.83x	21	0.86	0.91	25x	23x	1.4	1.3	1.3	1.1x	1.1x

Queries: Q_{desc} = All Descendants; Q_{child} = All Children; Q_{leaf} = All Leaves; $Q_{a\&d}$ = Check if Ancestor.
Parameters: Root–leaf = root-farthest-leaf pair; Deep parent–leaf = deepest parent-leaf pair (both true).
Siblings = shallow sibling pair; Distant leaves = leaf pair with greatest distance via their lowest common ancestor (both false).
Methods: *B* = Baseline; *D* = Dewey; *P* = PrePost; $S_D = B / D$; $S_P = B / P$. Runtimes: median of five runs in ms.
Graphs: F = forest; NT = truebase; DT = ultratall; WT = ultrawide; K = 1,000 nodes.
SNB/C, SNB/P, SNB/T = SNB SF1 trees; SNB = full SNB SF1 graph.
Rounding: runtimes and speedups to about two significant digits; K = 1,000 and M = 1,000,000. Bounds round outward.

Speedup color: 

gray = slowdown; yellow = 1x; green = speedup; darker green = larger speedup

Check if Ancestor (Query 11)

Graph	Query	Parameters	Apache AGE					Kuzu					Neo4j				
			B	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P
WT10	$Q_{a\&d}$	Root–leaf	10	1.3	1.3	7.5x	7.8x	18	0.91	0.88	20x	21x	1.8	1.1	1.1	1.6x	1.6x
WT10	$Q_{a\&d}$	Deep parent–leaf	10	1.3	1.3	7.5x	7.6x	19	0.85	0.97	22x	20x	1.3	1.2	1.2	1.1x	1.1x
WT10	$Q_{a\&d}$	Siblings	9.4	1.0	0.98	9.1x	9.6x	19	0.87	0.92	21x	20x	1.4	1.3	1.4	1.0x	1.0x
WT10	$Q_{a\&d}$	Distant leaves	9.7	0.99	0.97	9.8x	10x	18	0.91	0.92	19x	19x	1.5	1.3	1.3	1.1x	1.2x
WT100	$Q_{a\&d}$	Root–leaf	11	1.7	1.6	6.7x	7.1x	19	0.92	0.94	21x	20x	1.4	0.99	1.1	1.4x	1.3x
WT100	$Q_{a\&d}$	Deep parent–leaf	11	1.5	1.5	7.5x	7.6x	18	0.90	0.94	20x	19x	1.2	1.1	1.3	1.1x	0.98x
WT100	$Q_{a\&d}$	Siblings	9.8	1.2	1.3	8.3x	7.7x	18	0.92	0.93	19x	19x	1.3	1.1	1.1	1.1x	1.2x
WT100	$Q_{a\&d}$	Distant leaves	9.8	1.1	1.1	8.6x	8.8x	21	0.93	0.97	23x	22x	1.7	1.4	1.4	1.2x	1.2x
WT1K	$Q_{a\&d}$	Root–leaf	13	1.6	1.5	8.1x	8.3x	21	0.84	0.89	25x	23x	1.1	0.81	1.1	1.3x	0.94x
WT1K	$Q_{a\&d}$	Deep parent–leaf	12	1.5	1.5	7.8x	8.0x	19	0.85	0.89	22x	21x	1.3	1.3	1.3	1.0x	0.99x
WT1K	$Q_{a\&d}$	Siblings	11	2.6	2.5	4.2x	4.3x	21	0.88	1.1	24x	19x	1.4	1.4	1.4	1.1x	1.0x
WT1K	$Q_{a\&d}$	Distant leaves	11	2.6	2.5	4.1x	4.2x	23	0.94	0.90	24x	25x	1.6	1.4	1.4	1.1x	1.2x
WT10K	$Q_{a\&d}$	Root–leaf	19	1.7	1.8	11x	11x	21	0.86	0.86	25x	25x	1.4	1.2	0.91	1.2x	1.5x
WT10K	$Q_{a\&d}$	Deep parent–leaf	18	1.8	1.7	11x	11x	21	0.96	0.93	22x	23x	1.2	1.1	1.1	1.1x	1.1x
WT10K	$Q_{a\&d}$	Siblings	21	16	16	1.2x	1.3x	19	0.89	0.83	22x	23x	1.3	1.2	1.2	1.1x	1.1x
WT10K	$Q_{a\&d}$	Distant leaves	21	17	16	1.3x	1.3x	21	0.82	0.83	25x	25x	1.4	1.2	1.2	1.2x	1.1x
WT100K	$Q_{a\&d}$	Root–leaf	80	1.9	1.8	42x	44x	20	0.94	0.88	22x	23x	1.6	1.3	1.3	1.2x	1.2x
WT100K	$Q_{a\&d}$	Deep parent–leaf	81	1.8	1.8	44x	45x	23	0.84	0.89	27x	25x	1.3	1.3	1.3	1.0x	0.96x
WT100K	$Q_{a\&d}$	Siblings	130	160	160	0.78x	0.80x	20	0.88	0.91	23x	22x	1.2	1.3	1.4	0.96x	0.89x
WT100K	$Q_{a\&d}$	Distant leaves	130	160	160	0.79x	0.79x	20	0.88	0.93	22x	21x	1.4	1.4	1.5	1.0x	0.98x
SNB/C	$Q_{a\&d}$	Root–leaf	15K	2.0	1.9	7.3Kx	7.9Kx	22	1.0	0.93	22x	24x	1.6	1.2	1.1	1.3x	1.4x
SNB/C	$Q_{a\&d}$	Deep parent–leaf	15K	2.1	1.8	6.9Kx	8.0Kx	19	1.3	1.0	15x	19x	1.2	1.1	1.1	1.1x	1.0x
SNB/C	$Q_{a\&d}$	Siblings	16K	3.6K	3.5K	4.4x	4.5x	19	1.1	0.86	18x	22x	1.2	1.1	1.2	1.0x	1.0x
SNB/C	$Q_{a\&d}$	Distant leaves	16K	3.8K	3.7K	4.2x	4.3x	20	1.1	0.85	19x	24x	1.2	1.1	1.1	1.1x	1.1x
SNB/P	$Q_{a\&d}$	Root–leaf	15K	1.1	1.5	13Kx	9.7Kx	20	0.80	0.89	24x	22x	1.4	1.1	1.1	1.3x	1.3x
SNB/P	$Q_{a\&d}$	Deep parent–leaf	14K	1.6	1.5	9.2Kx	9.7Kx	17	0.88	0.91	20x	19x	1.3	1.1	1.1	1.2x	1.1x
SNB/P	$Q_{a\&d}$	Siblings	14K	3.2	3.2	4.5Kx	4.6Kx	21	0.87	0.92	24x	23x	1.2	1.1	1.2	1.1x	1.0x
SNB/P	$Q_{a\&d}$	Distant leaves	14K	3.2	3.1	4.5Kx	4.6Kx	18	0.88	0.90	20x	20x	1.3	1.2	1.2	1.1x	1.1x
SNB/T	$Q_{a\&d}$	Root–leaf	15K	1.6	1.7	9.0Kx	8.6Kx	19	0.82	0.95	23x	20x	1.2	1.1	1.2	1.1x	1.0x
SNB/T	$Q_{a\&d}$	Deep parent–leaf	14K	1.4	1.7	10Kx	8.6Kx	21	0.83	0.88	25x	23x	1.4	1.1	1.1	1.3x	1.3x
SNB/T	$Q_{a\&d}$	Siblings	14K	1.1	1.0	13Kx	14Kx	19	0.83	0.89	23x	22x	1.2	1.1	1.2	1.1x	1.0x
SNB/T	$Q_{a\&d}$	Distant leaves	14K	1.0	0.99	14Kx	14Kx	18	0.85	0.82	22x	22x	1.3	1.5	1.2	0.87x	1.1x

Queries: Q_{desc} = All Descendants; Q_{child} = All Children; Q_{leaf} = All Leaves; $Q_{a\&d}$ = Check if Ancestor.
Parameters: Root–leaf = root-farthest-leaf pair; Deep parent–leaf = deepest parent-leaf pair (both true).
Siblings = shallow sibling pair; Distant leaves = leaf pair with greatest distance via their lowest common ancestor (both false).
Methods: *B* = Baseline; *D* = Dewey; *P* = PrePost; $S_D = B / D$; $S_P = B / P$. Runtimes: median of five runs in ms.
Graphs: F = forest; NT = truebase; DT = ultratall; WT = ultrawide; K = 1,000 nodes.
SNB/C, SNB/P, SNB/T = SNB SF1 trees; SNB = full SNB SF1 graph.
Rounding: runtimes and speedups to about two significant digits; K = 1,000 and M = 1,000,000. Bounds round outward.

Speedup color: 

gray = slowdown; yellow = 1x; green = speedup; darker green = larger speedup

LDBC SNB SF1: IC12, IS2, IS6

Graph	Query	Parameters	Apache AGE					Kuzu					Neo4j				
			B	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P
SNB	IC12	–	190K	170K	170K	1.1x	1.1x	400	300	400	1.3x	0.98x	160	120	110	1.3x	1.5x
SNB	IS2	–	61K	3.8K	13K	16x	4.7x	44	240	280	0.18x	0.16x	4.5	4.5	11	0.99x	0.42x
SNB	IS6	–	17K	2.2K	3.3K	8.0x	5.3x	29	23	120	1.3x	0.24x	1.9	2.0	1.8	0.95x	1.0x

Queries: IC12 = Interactive Complex 12; IS2 = Interactive Short 2; IS6 = Interactive Short 6.

Methods: *B* = Baseline; *D* = Dewey; *P* = PrePost; S_D = *B* / *D*; S_P = *B* / *P*. Runtimes: median of five runs in ms.

Graphs: F = forest; NT = truebase; DT = ultratall; WT = ultrawide; K = 1,000 nodes.

SNB/C, SNB/P, SNB/T = SNB SF1 trees; SNB = full SNB SF1 graph.

Rounding: runtimes and speedups to about two significant digits; K = 1,000 and M = 1,000,000. Bounds round outward.

Speedup color: 

gray = slowdown; yellow = 1x; green = speedup; darker green = larger speedup